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(b) Adaptive radiation

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HF Osburn coined the term adaptive radiation. He defined it as great variety of adaptations to different eco-niches.

Simpson (1950) defined it as proliferation of new species from single ancestral group.

Factors affecting adaptive radiation

adaptive potential
of species

adaptive opportunities
provided by environment.

Adaptive radiation happens when a species fits into an environment into which earlier it ~~was~~ didn't adapted. This is caused by certain factors.

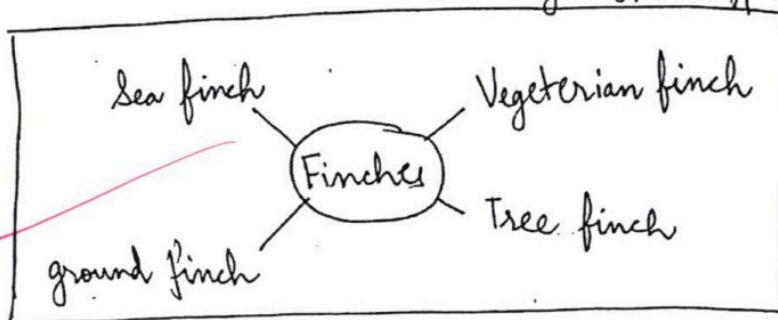
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Darwin's finches

Long ago, a specie of finch migrated to gallapagos islands. Due to adaptive radiation, beaks of finches adjusted according to diet available on island

Fig: different types of finches



Significance

- 1) It helps in evolutionary studies
- 2) Understanding effect of environment on evolution of species.

attempted well

Hence, adaptive radiation plays an important role in evolution and adjustment of species to environment.